

Exploiting ViewState Using ysoserial.net

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Content

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[]()Intro

This writeup was inspired by the following [blog](#) that discusses .NET ViewState Exploitation. In this writeup, I explored the TypeConfuseDelegate and ActivitySurrogateSelectorFromFile [ysoserial.net](#) gadgets, the gadgets most commonly used to exploit ViewState and gain RCE.

The writeup also slightly touches upon the concept of COM and .NET Interoperabilities that serves as a foundation of the gadgets, and a lot of .NET documentations. Enjoy reading!

[]()ViewState Basics

[]()MAC Enabled=true + Invalid ViewState

[]()About ViewStateUserKey

- Used to prevent CSRF attacks.
- `ViewStateUserKey` is typically set to a value such as the current user's username or the user's session identifier.
- Sample scenario:
- A client has two tabs open in a browser.
- Tab 1: Logs in as user A, `__VIEWSTATE` with A's `ViewStateUserKey` is rendered.
- Tab 2: Client logs out, logs in as User B, goes back to Tab 1 and submit the form.

- Submitted `ViewStateUserKey` is A's, but on the server's side is B's, mismatch causing error.

ViewState References:

- <https://www.c-sharpcorner.com/UploadFile/225740/what-is-view-state-and-how-it-works-in-Asp-Net53/>
- <https://www.c-sharpcorner.com/uploadfile/37db1d/looking-deep-inside-postback-and-viewstate-in-Asp-Net-3-5/>

[(())TypeConfuseDelegate

This payload successfully writes a file, but shows 500 Internal Server Error Hence, we might not verify whether the file is successfully written.

Take Note:

- It calls `HiddenFieldPageStatePersister` Class
- Inheritance: `Object` -> `PageStatePersister` -> `HiddenFieldPageStatePersister`

[(())ActivitySurrogateSelectorFromFile

[(())Passing .cs and supporting .dll files

Error: Unable to cast object of type 'State' to 'System.Web.UI.Pair' What this indicates:

- The top-level object passed in the ViewState parameter is of type 'State'
- [ASP.NET](#) ViewState expects to work with 'System.Web.UI.Pair'

[(())Passing .dll from a Compiled .cs

Convert the .cs file to .dll

Pass the converted payload to ysoserial

[(())Why one works and the other does not?

Two main questions:

- Does passing a .cs file along with supporting .dll unable to result in a Pair object?
- If so, does this fail because of:
 - [ASP.NET](#) ViewState deserialization expects a Pair object, or
 - ActivitySurrogateSelectorFile expects to work with Pair objects?

[(())Analyzing ActivitySurrogateSelectorFile

The `Generate()` function in `ActivitySurrogateSelectorFromFileGenerator.cs` calls `GadgetChainsToBinaryFormatter()` from `ActivitySurrogateSelectorGenerator.cs`

The `GadgetChainsToBinaryFormatter()` instantiates a new object from `GadgetChains()`
`GadgetChains()` itself makes use of `Hashtable`, `IEnumerator`, etc.

`Hashtable` itself is a collection of key/value pairs, will it have anything to do with the errors? Not necessarily.

Referencing this article: <https://googleprojectzero.blogspot.com/2017/04/>

- `ActivitySurrogateSelector` does make use of `hashtable` to trigger `IEnumerator` so our payload gets deserialized, but that happens before we pass the final payload to `ViewState`.
- If an error about `Pair` occurs, it's due to the top-level object in the serialized payload not being a `Pair`, which only becomes a problem at deserialization time (on the server).

[(())Reason #2: The gadget chain behavior itself

`ActivitySurrogateSelector` does not necessarily expect to work with `Pair` Objects. See `ActivitySurrogateSelector` analysis for detailed walkthrough.

[(())Analyzing `ASP.NET ViewState` Handler

As far as I know, `ViewState` deserialization uses `ObjectStateFormatter()`, making it able to deserialize any objects (?)

Further analyzing how `ViewState` works, apparently it uses

`LoadPageStateFromPersistenceMedium`, which uses `PageStatePersister.Load`

Recall that when using `TypeConfuseDelegate`, an error message about the `HiddenFieldPageStatePersister` occurs

Taking a look at the `PageStatePersister.Load()` Method, we see a deserialization in action, which will cast the result into `Pair`

Few things to note:

- `PageStatePersister.Load()` is called to load `ViewState` from a persistence medium.
- It uses an `IStateFormatter` to deserialize `ViewState` data
- The deserialized object is casted to `Pair` to initialize the `ViewState` and `ControlState` property
- The pattern will be: `Pair(pageViewState, controlState)`

Some few information from .NET documentation:

By default, `ASP.NET` page state is serialized and deserialized by an instance of the `ObjectStateFormatter` class; however, site and adapter developers can implement the `IStateFormatter` interface on their own types to perform this work.

About the `ObjectStateFormatter` Class:

Other types not listed above are binary-serialized using a `BinaryFormatter` object if they implement the `ISerializable` interface or are decorated with the `SerializableAttribute` attribute.

Notice that we are deserializing a class (not listed in the list stated in [reference](#)). So our object will be deserialized by `BinaryFormatter`.

Which makes sense, since `ysoserial ActivitySurrogateSelector` gadget itself serialized the payload using `BinaryFormatter`

Conclusion:

- The error: `Unable to cast object of type 'State' to 'System.Web.UI.Pair'` happens when casting the deserialized object
- When passing a `.cs` file, `ysoserial` compiles it and wraps it in a `State` object, which becomes the top-level serialized instance — breaking the [ASP.NET ViewState](#) assumption.
- Execution path:

- `Page.LoadPageStateFromPersistenceMedium()`
- `PageStatePersister.Load()`
- `IStateFormatter.Deserialize()` → Expects: `Pair` object
- This has nothing to do with the internals of the gadget itself (the early assumption of `SortedKey`'s and `Hashtable`'s relevance to `Pair` objects)
- `ObjectStateFormatter` takes many types of data structures, the `Pair` fails because of the type casting

Confusion: Why `.dll` works, `.cs` does not?

- When you pass a compiled `.dll`, `ysoserial` loads the class from the DLL, instantiates it, and plugs the object into the `ActivitySurrogateSelectorFromFile` gadget chain. This chain is then wrapped in a `System.Web.UI.Pair`, as required by `ViewState` deserialization.
- `ysoserial.exe` cannot compile the `.cs` file on its own — it expects a DLL, not source code. So if it works at all when you pass a `.cs` and `.dll`, it must be because: The DLL is doing the real work, or the `.cs` was previously compiled, cached, or accidentally referenced through a different mechanism.

[(())Analyzing The Gadgets

[(())Background Knowledge

[(())Managed and Unmanaged Code

- Managed code: execution is managed by a runtime (CLR: compile managed code to machine code and executing it)
- C#, [VB.NET](#)

- Unmanaged code: Programmer manages memory to security considerations
- C++, VBG

COM (unmanaged) vs. .NET (managed)

- Object Lifetime
- COM: Client manages when to free/release the object.
- .NET: CLR automatically cleans up objects with GC.
- Discovering Capabilities
- COM: Requests interface, receives interface pointer if available.
- .NET: Use reflection to see all properties, methods, and interfaces of the objects.
- Memory Management
- COM: Expects the object to stay in one place, no mechanism to deal with dynamic object location.
- .NET: Objects reside in memory managed by .NET runtime execution environment, moves in memory for performance reasons.

COM and .NET Interoperabilities

Interoperability: Passing the code between managed and unmanaged boundaries.

- .NET might not be COM, but should be able to interoperate with COM
- .NET can implement and consume COM objects
- These interoperabilities are supported by Runtime Callable Wrapper (RCW) and COM Callable Wrapper (CCW)

Sample: Out-of-Process COM Server in C

Server Implementation

This is a .NET Object

Client Implementation

- Unmanaged COM client (C++)

Requests interface and receives interface pointer (if available).

- Managed client (C# via COM Interop)

Use System.Reflection to see all information about the loaded assemblies (the objects).

[(())How CCW Works As A Translator

- When COM client calls a .NET object, a CCW is needed as a translator between COM and .NET.
- The CCW is automatically created by the .NET runtime.
- CCW makes the .NET object look like a native COM object to the COM client.

To Query Interface

- Instantiates a COM object using its CLSID, requests the object to return a pointer to IUnknown interface, pUnknown will point to the COM's IUnknown vtable
- Now having pointer to IUnknown, use QueryInterface to get IDispatch pointer (pDisp)

[(())Now What If The Caller is .NET Client?

Simply put, when .NET runtime gets hold of a COM object, it will go through a "process" to determine if it can unwrap the object from its CCW and avoid creating an RCW. The process is illustrated above It might end up calling BinaryFormatter::Deserialize()!

[(())Attacker's Perspective (Local Privilege Escalation)

Make the server to try and create an RCW, for a serializable .NET object that is exposed over COM (the above inner box containing ".NET Object").

This abuses WMI

The server itself is not vulnerable, until it acts as a DCOM client: retrieving and deserializing attacker-controlled data.

Trick:

- Pass a .NET COM object to the server's `Equals` method
- The runtime tries to convert to RCW
- Runtime checks if it's a CCW wrapped .NET object, resulting in calling `GetSerializedBuffer`
- Send a serialized `Hashtable` to the server containing delegate object as one of the keys.
- The server contains COM implementation of the `IHashCodeProvider` Interface
- Custom deserialization code causes `IHashCodeProvider::GetHashCode` to be called, rebuilding the internal hash structures, running over each keys.
- The function meets the Delegate object (which is serializable), so it will gets passed back to the client (refer to the diagram above)
- The client is written in native code, so `IManagedObject` automatic serialization won't occur.

- The Delegate object is in the server while we are exposed to the CCW.
- Make a call via CCW to start a new process with server's privileges.

The .NET COM Server becomes a DCOM client when it calls a method (GetHashCode) on a COM Interface, which is implemented by a remote object controlled by the attacker. The serialized hashtable contains the COM implementation (not .NET!!!) of the IHashCodeProvider, causing the .NET COM Server to be a DCOM Client when calling the method. Works in local privilege escalation but not guaranteed for RCE.

[[()TypeConfuseDelegate

[[()Key Idea

Creates a `String.Compare` delegate, multicasts it to later create type confusion, then uses this multicast delegate as a comparer in a `SortedSet` Object. Sets the attacker's payload inside the `SortedSet`, then modifies the delegate's invocation list to call `Process.Start()`.

[[()How It Works

[[()Step 1: Get user's input arguments

[[()Step 2: Create a Multicast Delegate

Multicast delegate contains a list of the assigned delegates. When the multicast delegate is called, it invokes the delegates in the list in order. The types must match first so it will be valid in the `ComparisonComparer`, afterwards we can modify the Invoke List 1 to `System.Process.Start`.

[[()Step 3: Create a ComparisonComparer instance with the delegate

`ComparisonComparer` performs a case-sensitive comparison of two objects with the same type. Hence, since the multicast delegate contains two same types, it is valid.

[[()Step 4: Add user's arguments to the SortedSet

[[()Step 5: Modify the multicast delegate's list of invocation

[[()Why specifically modify `_invocationList[1]` ?

- Upon object construction, `DelegateSerializationHolder.GetRealObject()` will be called.
- The above function treats the first element of the delegate list (`_invokeList[0]`) as the type for the entire `MultiCastDelegate` object.

- We want to keep the type `Comparison<string>` .

[(())ActivitySurrogateSelector

[(())Key Idea

Generates an enumerable (hashtable) via `LINQ` , serializes it using `ObjectSurrogate` , forces enumeration to trigger deserialization via the `ToString()` method.

[(())How It Works

[(())Step 1: Hashtable construction

-

Create the gadget chains in the form of `IEnumerable`

-

Create a new instance of `DesignerVerb` and `IDictionary`

```
var list = new List<IEnumerable>();
var dict = new Dictionary<string, object>();
```

`DesignerVerb` and `IDictionary` is used hand-in-hand to trigger `ToString()` Will result in the `LINQ` enumerator being walked

-

Initialize a normal empty list.

-

Populate the list with the pre-loaded gadget chains.

```
list.Add(new DesignerVerb("Gadget Chain", () => {
    // ... gadget chain logic ...
}));
```

This becomes a list of `IEnumerable` elements.

-

Initialize a Hashtable

-

Replace the Hashtable's existing keys with `DesignerVerb`

```
dict.Clear();
dict.Add(new DesignerVerb("Gadget Chain", () => {
    // ... gadget chain logic ...
}), null);
```

So `ToString()` method could be triggered since the key is not a string

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Add the modified Hashtable to the list

[(())Step 2: Serialize the list

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List serialization

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And serialize our input arguments passed via ysoserial

The result is what we will pass to the server.

[]()Step 3: Server receives the payload

- The Hashtable will get deserialized and all keys will be rehashed.
- If two same keys are found upon rebuilding, an exception will be thrown.
- The exception will end up calling `GetResourceString` and attempt to format the keys.
- If the key is not a `String` type (since we modified it), it will trigger the `ToString()` method, firing up our gadget chain.

[]()Summary

- Delegates: A type that represents references to methods with a particular parameter list and return type.
- We create an open delegate, which not bound to an instance and needs to be passed explicitly at invocation.
- In the end, each `Type` in `e5` is passed to `Activator.CreateInstance(Type)`

[]()References:

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